

ZAVKHAN, Mongolia

WMO: 442160

Lat: 48.8239N

Lon: 93.0983E

Elev: 939

StdP: 90.54

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.9	-36.8	-41.4	0.1	-38.7	-39.4	0.1	-36.8	9.8	-16.2	8.0	-19.9	1.8	320	0.639

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	32.0	16.3	30.3	15.8	28.8	15.3	17.7	29.0	16.8	27.5	16.2	26.5	3.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.7	11.0	21.8	12.6	10.2	21.3	11.5	9.5	21.7	54.0	29.5	51.0	27.9	48.9	26.8	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.2	10.1	8.8	-38.9	35.4	4.6	1.7	-42.2	36.6	-44.9	37.6	-47.5	38.6	-50.8	39.8		
			-39.3	18.7	4.5	1.4	-42.6	19.7	-45.2	20.5	-47.8	21.3	-51.0	22.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.5	-26.8	-21.9	-5.6	7.2	13.5	20.4	22.7	20.1	13.0	3.8	-9.0	-20.8
	DBStd	17.79	5.70	7.43	7.72	5.15	4.68	3.52	2.86	3.40	4.49	4.74	5.61	5.91
	HDD10.0	4390	1139	894	483	111	21	0	0	0	22	196	569	955
	HDD18.3	6456	1398	1127	742	334	159	20	3	22	168	449	819	1213
	CDD10.0	1290	0	0	1	26	129	312	393	313	111	4	0	0
	CDD18.3	314	0	0	0	0	9	82	138	77	7	0	0	0
	CDH23.3	2916	0	0	0	6	141	775	1252	652	90	0	0	0
	CDH26.7	892	0	0	0	0	28	229	432	187	16	0	0	0
Wind	WSAvg	3.2	1.6	2.0	3.2	4.2	4.5	3.9	3.9	3.7	3.6	3.3	2.7	1.9
Precipitation	PrecAvg	186	5	4	6	11	15	27	45	30	15	9	11	7
	PrecMax	295	10	7	14	22	26	56	121	66	36	22	24	14
	PrecMin	106	1	2	0	2	4	11	11	8	5	2	4	2
	PrecStd	45	2	1	3	5	6	10	24	15	7	5	5	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-10.7	-1.3	14.7	24.0	29.7	33.4	35.1	33.0	28.5	19.0	6.7	-4.4
		MCWB	-12.3	-5.1	5.3	11.8	14.2	17.1	17.9	16.1	14.0	8.7	1.8	-7.3
	2%	DB	-14.0	-5.4	10.9	21.2	26.9	31.1	32.7	30.5	26.1	16.0	3.4	-8.7
		MCWB	-15.0	-8.2	3.1	9.7	12.8	15.5	17.0	16.1	13.1	7.2	-0.7	-10.1
	5%	DB	-16.6	-8.5	8.1	18.9	24.8	29.4	30.7	28.7	23.5	13.7	1.0	-11.1
		MCWB	-17.4	-10.4	1.7	8.5	11.6	14.8	16.4	15.4	12.3	5.8	-2.2	-12.1
	10%	DB	-18.8	-11.5	5.3	16.4	22.6	27.7	29.1	27.0	21.1	11.6	-1.3	-13.0
		MCWB	-19.3	-12.8	0.1	7.0	10.6	14.5	15.9	14.9	11.3	4.7	-4.1	-13.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-12.1	-5.2	6.0	12.3	15.2	19.5	19.4	17.8	15.8	9.1	2.1	-6.9
		MCDB	-10.7	-1.6	12.9	23.1	27.3	30.8	30.8	29.2	27.1	18.2	6.2	-3.9
	2%	WB	-15.1	-8.1	3.4	10.1	13.3	16.7	18.2	16.8	13.8	7.4	-0.4	-10.2
		MCDB	-14.1	-5.5	10.1	20.7	25.5	27.6	29.7	27.5	22.7	15.0	2.8	-8.8
	5%	WB	-17.3	-10.4	1.8	8.6	12.1	16.0	17.3	16.2	12.6	6.2	-2.2	-12.1
		MCDB	-16.6	-8.5	8.1	18.1	24.1	26.5	28.3	26.3	22.6	13.2	1.0	-11.2
	10%	WB	-19.3	-12.8	0.0	7.3	11.0	15.2	16.6	15.5	11.6	4.9	-3.9	-13.8
		MCDB	-18.8	-11.5	4.9	15.7	21.8	25.7	27.1	25.3	20.5	11.3	-1.3	-13.0
Mean Daily Temperature Range	5% DB	MDBR	8.3	10.6	13.5	13.5	12.6	11.4	10.7	11.0	12.3	12.2	9.1	7.5
		MCDBR	10.0	12.4	15.7	15.9	15.6	13.7	13.2	13.3	15.3	14.6	10.9	8.1
		MCWBR	9.2	10.3	10.0	8.9	7.8	5.7	4.9	5.0	6.9	8.5	8.0	7.2
	5% WB	MCDBR	9.9	12.4	15.0	15.2	14.5	11.3	11.5	11.6	14.2	14.0	10.8	7.9
		MCWBR	9.2	10.3	9.8	8.9	7.7	5.1	4.9	4.8	6.6	8.5	7.9	7.0
Clear-Sky Solar Irradiance	taub	0.254	0.282	0.327	0.368	0.369	0.374	0.372	0.352	0.317	0.298	0.275	0.254	
	taud	2.302	2.248	2.221	2.223	2.265	2.341	2.381	2.408	2.456	2.433	2.320	2.268	
	Ebn at Noon	819	870	881	877	890	886	882	886	889	847	779	759	
	Edh at Noon	69	94	116	129	129	121	114	106	91	78	68	63	
All-Sky Solar Radiation	RadAvg	1.35	2.42	4.06	5.76	6.99	7.16	6.85	5.97	4.85	3.18	1.69	1.08	
	RadStd	0.15	0.18	0.28	0.22	0.34	0.24	0.34	0.25	0.15	0.12	0.18	0.16	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	-2.14	-2.66	N/A	-0.90	-1.19	N/A	N/A	N/A	N/A

Nomenclature: See separate page